This assignment will not be graded and is only for practice.

Level 1

1) If $||a|| = 3$, $||b|| = 2\sqrt{2}$ and angle between a and b is 45°, then the value of $a.b$ is

No, the answer is incorrect.
Score: 0

Accepted Answers:

(Type: Numeric) 6

1 point

2) Given that $\|a\| = 3$ and $\|b\| = 5$ and $a \cdot b = 7.5$, for two vectors a and b in R^n. Find the angle (in degrees) between the two vectors a, b. Note: $\cos 60^\circ = 1/2, \cos 30^\circ = \sqrt{3}/2$.

1 point

- ○ 30°
- ○ 60°
- ○ 120°
- ○ 150°

No, the answer is incorrect.
Score: 0

Accepted Answers:

60°

3) Consider two vectors $a = (3, 4, -1), b = (2, -1, 2)$ in R^3. Choose the correct options.

1 point

- ☐ Length/Norm of a is $\sqrt{26}$ and that of b is 3.
- ☐ Length/Norm of a is 5 and that of b is 3.
- ☐ Length/Norm of a is $\sqrt{24}$ and that of b is $\sqrt{7}$.
- ☐ Angle between the two vectors is 90 degrees.

☐ Angle between the two vectors is approximately 0 degrees.

No, the answer is incorrect.

Score: 0

Accepted Answers:

Length/Norm of a is $\sqrt{26}$ and that of b is 3.
Angle between the two vectors is 90 degrees.

4) If the dot product of two vectors is zero, then choose the correct option. ***1 point***

- ◯ The two vectors are perpendicular to each other.
- ◯ The two vectors are parallel to each other.
- ◯ The two vectors are exactly the same.
- ◯ The length of the two vectors must be the same.

No, the answer is incorrect.

Score: 0

Accepted Answers:

The two vectors are perpendicular to each other.

Level 2

5) If $a = (6, -1, 3), b = (4, c, 2)$ where $a, b \in \mathrm{R}^3$ and a and b are perpendicular to each other, then find the value of c?

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 30

1 point

6) Consider two vectors $a = (1, 2), b = (2, 2)$ and θ is the angle between them. The sum of the two vectors is given by $c = a + b$. Choose the correct options. ***1 point***

- [ ] Length/Norm of c is 5.
- [ ] Length/Norm of c is 25.
- [ ] Length/Norm of c is 4.
- [ ] $\cos\theta = \frac{3}{\sqrt{10}}$.
- [ ] $\cos\theta = \frac{3}{\sqrt{5}}$.

No, the answer is incorrect.

Score: 0

Accepted Answers:

Length/Norm of c is 5.
$\cos\theta = \frac{3}{\sqrt{10}}$.

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7) Consider 3 vectors a, b, c in R^3 and a scalar λ in R. Choose the set of correct options. Note: $(.)$ represents the dot product. ***1 point***

- [ ] $\lambda(a \cdot b) = (\lambda a) \cdot b$
- [ ] $\lambda(a \cdot b) = (\lambda b) \cdot a$
- [ ] $\lambda(a \cdot b) = (\lambda a) \cdot (\lambda b)$
- [ ] $(a + c) \cdot b = a \cdot b + c \cdot b$
- [ ] $(a + c) \cdot b = a \cdot c + c \cdot b$
- [ ] $a \cdot a = 0, b \cdot b = 0$ if and only if a, b are null vectors.

No, the answer is incorrect.

Score: 0

Accepted Answers:

$\lambda(a \cdot b) = (\lambda a) \cdot b$
$\lambda(a \cdot b) = (\lambda b) \cdot a$
$(a + c) \cdot b = a \cdot b + c \cdot b$

$a \cdot a = 0, b \cdot b = 0$ if and only if a, b are null vectors.

8) Consider two vectors $a = (1, 3, 5, 7, 9)$ and $b = (2, 4, 6, 8, 10)$. Choose the set of correct options. ***1 point***

- [ ] The length of b is more than a.
- [ ] The length of a is more than b.
- [ ] The length of $(a - b)$ is $\sqrt{5}$.
- [ ] The length of $(a - b)$ is $\sqrt{50}$.

No, the answer is incorrect.

Score: 0

Accepted Answers:

The length of b is more than a.
The length of $(a - b)$ is $\sqrt{5}$.

Your score is: 0/8